

Fruit Fly

Team: Cheesecake

Element C: Presentation and Justification of Solution Design Requirements

Our boarding school residents need a way to remove fruit flies from their dormitories. This is because residents are irritated by the fruit flies and they could also pose a health risk. The fruit flies are attracted to a food item that was left in the dormitories a few weeks ago. Fruit flies have a short reproduction time so the quicker they are removed the easier it will be because there are less flies to remove. Quality of life in the dormitories would also improve if the fruit flies were to be removed.

Removing at least 80% of the flies is necessary because the method we use needs to be effective in its goal. Our design needs to be intuitive because even though it is necessary to make life easier, it should not make your life harder by being difficult to assemble and take apart. Our design can not break if it is being touched or moved because the design would be ineffective if it is broken. The reason why our design should not require high maintenance is because we do not want the product to stop one inconvenience by creating another one, which is taking a high amount of time to maintain the prototype. Our design cannot have too much of an odor because an odor would create another inconvenience in the area with the flies. Our prototype needs to be smaller than $1,000 \text{ cm}^3$ because if our design takes up too much space it will be at risk of falling and the large size will make the room more cluttered. We decided on the price being at most \$20 because our budget is limited. Our design can not be temperature sensitive because the dormitories have high temperature fluctuation.

Design requirements based upon importance:

- a. Remove 80% of flies (High)
- b. Not have an odor at all (High)
- c. 1,000 cm³ maximum size (High)
- d. Price - \$20 maximum (Medium)
- e. Easy to use (Medium)
- f. Will not break if touched or moved (Medium)
- g. Maintain one or less times per week (Low)
- h. Cannot be fragile to temperature changes (Low)

Our stakeholders agreed with the design. We do not need to alter our list.